Team Name: _____ and #: B_____
Score: /100

TSSO INVITATIONAL

Dynamic Planet KEY

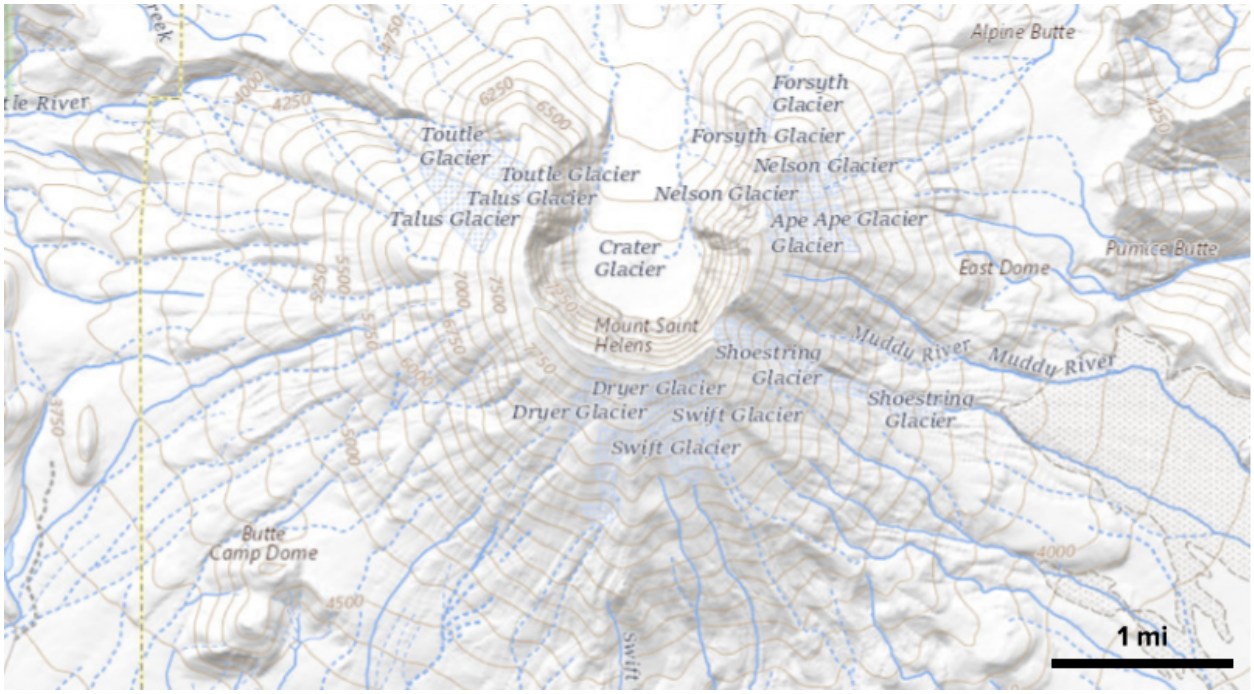


Tie breaker(s): 29, 42, 67, 51, 20
Time allotted: 50 minutes

Instructions:

1. You will have 50 minutes to complete this test.
2. This test is 75 questions and is worth 100 points. You should be able to complete it within the time period.
3. There are two equally weighted sections, Image Interpretation and Case Studies.
4. Point values for multiple select questions are not necessarily equal to the number of correct answers.
5. Tiebreakers are defined above and marked in the test. If two teams score equally overall and on every tiebreaker, their answer quality will be used to break a tie.
6. Good luck!

Section 1: Image Interpretation



A topographic map of Mount St. Helens with freshwater features labeled. (Please excuse the image quality, or lack thereof.)

1. What drainage pattern does this region have?
    a. Dendritic
    b. Radial
    c. Centripetal
    d. Parallel
    e. None of the above

2. Notice the Muddy River extending east of the mountain. Why would a river fed by glaciers be so muddy? (2)

    Glaciers generally have a lot of sediment due to erosion (2), so their meltwater typically does as well.

3. Using the scale in the bottom right, estimate the gradient of the first mile of the Muddy River's main channel in ft/ft. (1 mile = 5280 ft)

    0.47 (10 contours per mile), accept anywhere from 0.4-0.55

4. Based on the information above, what type of river would Muddy River likely be once it flattened out? (2)
    a. Straight

b. Meandering
c. Braided
d. Anastomosing
e. None of the above

5. In the eruption of 1980, many of Mount St. Helens' glaciers melted, creating a flow of water mixed with ash called what?
    a. Lahar

6. Notice the aptly named Crater Glacier in the volcano's crater. If this volcano were to erupt as it did in 1980, would this glacier likely make it more or less dangerous? Why? (2)
    a. More (1) because it would increase the likelihood of lahars, which are extremely destructive and one of the most dangerous parts of an eruption (1).

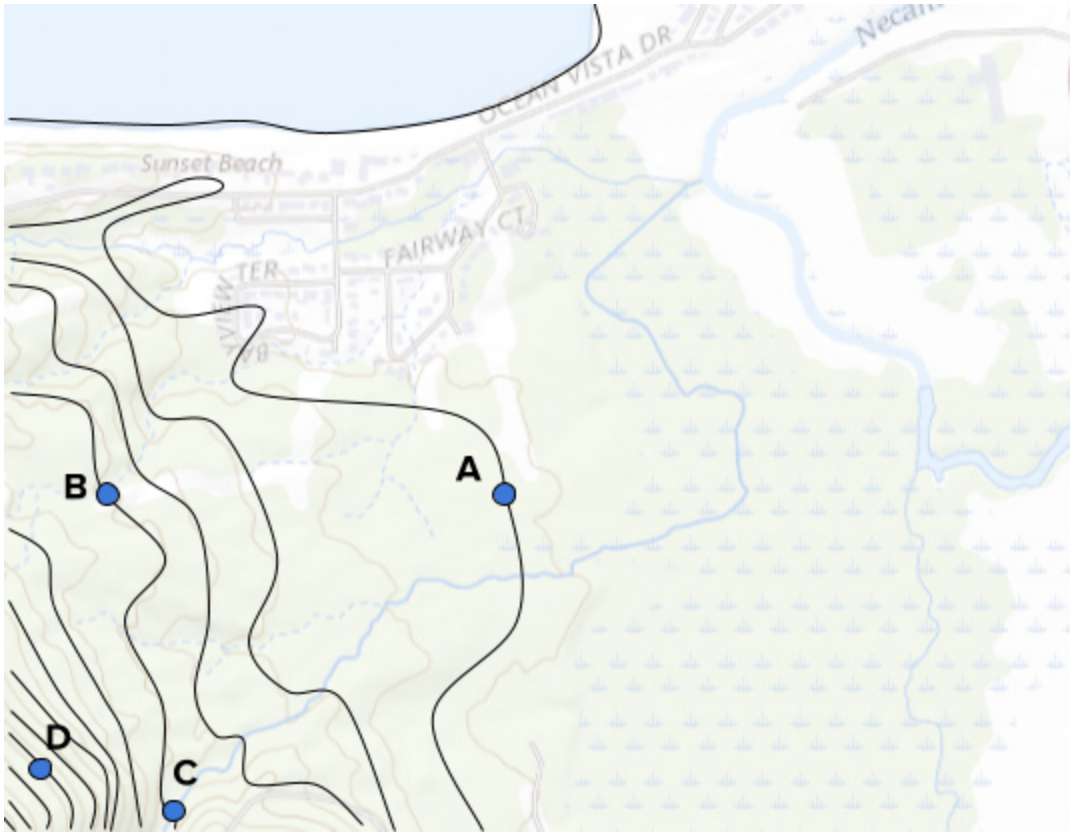


A topographic map of part of Seaside, WA with freshwater features labeled.

7. Which of the following best describes the land on the right side of the image?
    a. Tidal flat

b. **Wooded marsh**
c. Intermittent lake
d. Mangrove
e. None of the above

8. Why does this feature not extend to the region of higher elevation on the left?
    a. Marshes need a water table very close to the surface (1). However, as elevation increases, the water table typically gets farther from the surface.



Water table contours for this region overlaid above the original map. Both the original and new contours represent 100 ft intervals.

9. At which point is the water table highest?
    a. A
    b. B
    c. C
    d. D

10. At which point is the water table closest to the surface?
    a. A
    b. B
    c. C

d. D

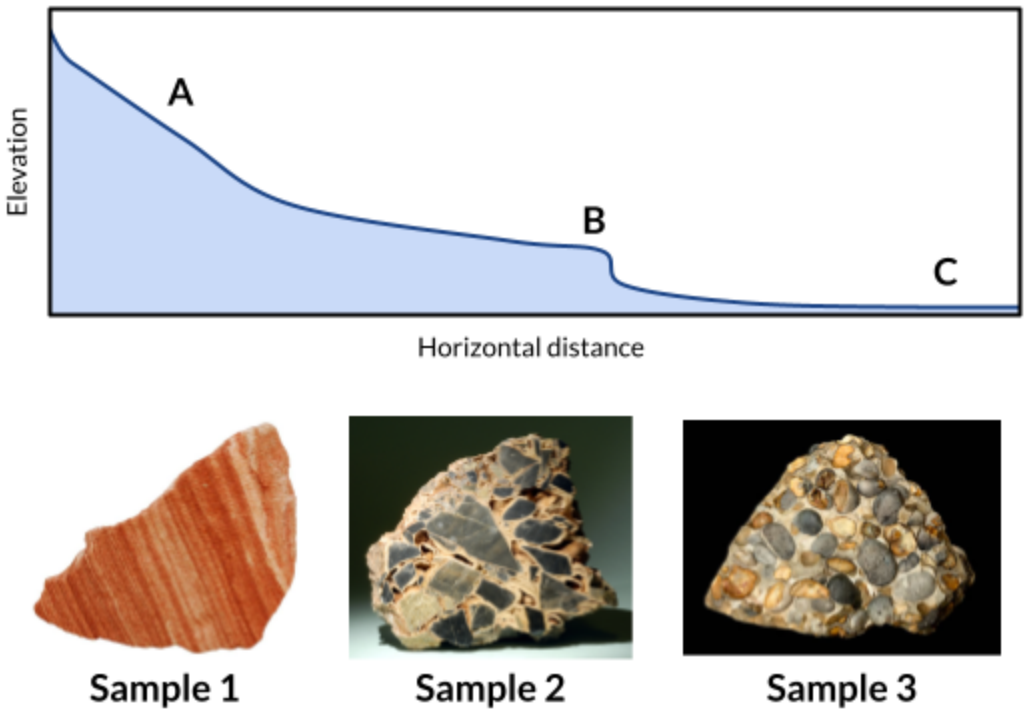
11. Which point would likely be the most eroded?
    a. A
    b. B
    c. C
    d. D

12. Which point would likely have the most alluvial sediment?
    a. A
    b. B
    c. C
    d. D

13. A geologist drills a well at point D, expecting to find water at a depth of 200 ft. However, after 50 ft, she finds a small body of water. She has likely encountered a(n) ___ aquifer.
    a. Perched

14. The geologist finds the main aquifer at a depth of 200 ft as expected, but once the well hits the aquifer, the water level rises on its own to 100 ft. By doing this, she has created a(n) ___ well.
    a. Artesian

15. Which of the following types of sediment did the geologist most likely have to drill through before reaching the main aquifer? (2)
    a. Clay
    b. Gravel
    c. Sand
    d. Silt



A diagram of a stream's elevation graphed with distance, with three rock samples.

16. What is this representation of a stream typically called?
    a. Stream profile

17. The stream gradient flattens out as it gets closer to closer to the bottom line of the rectangle. This means the bottom of the rectangle is approximately equal to what level?
    a. Base level (0.5 for sea level)

18. How would sinuosity at A compare to at C?
    a. Higher than C
    b. Less than C
    c. About the same as C

19. What is the general name for features such as the waterfall at B where the stream's slope abruptly changes at a single point?
    a. Knickpoints

20. Why are features like the waterfall at B relatively rare in mature streams? (2)
    a. This abrupt change in slope typically evens itself out (1). Erosion occurs on the top and deposition occurs on the bottom (1).

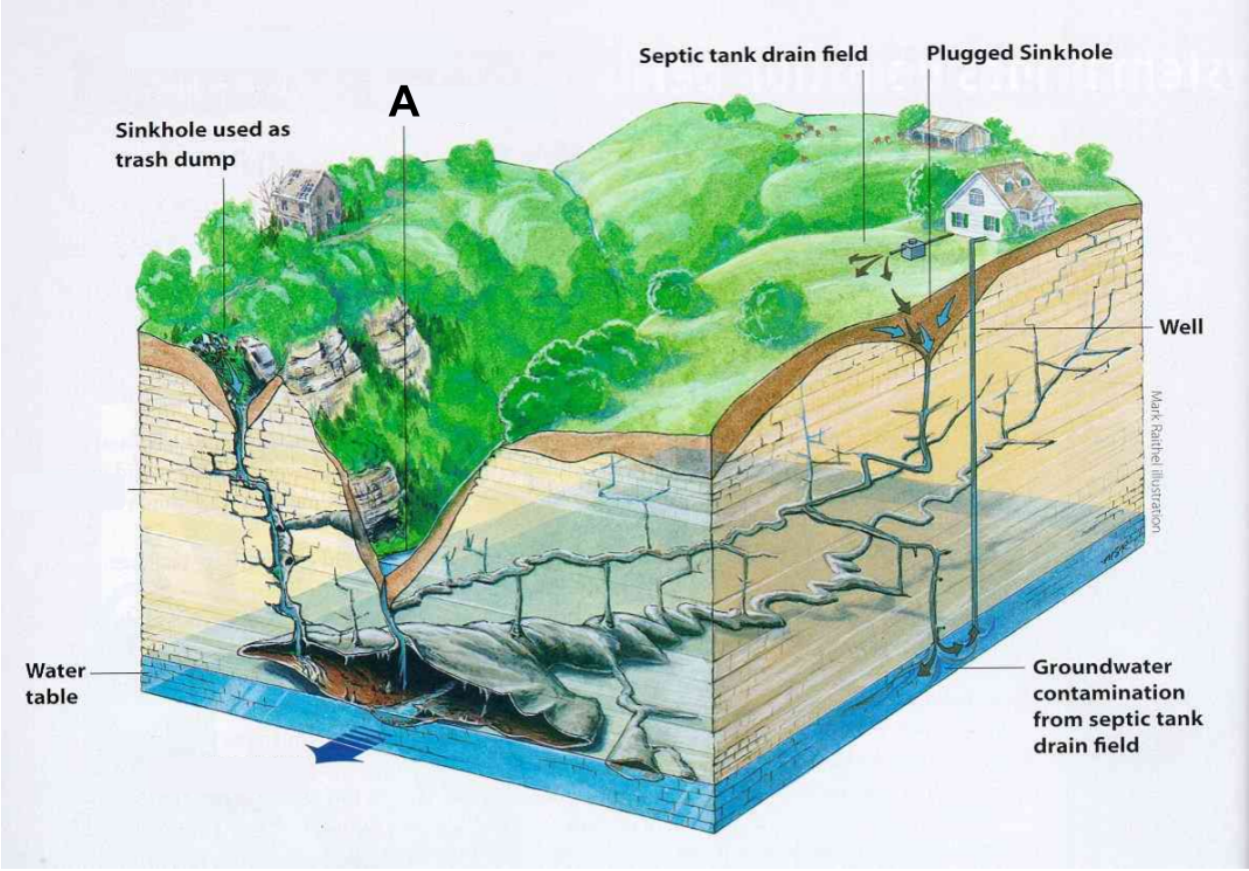
21. Propose a possible method of formation for the abrupt change in slope at B.

a. Fault, drop in sea level, change in bedrock type… (accept plausible answers)

22. Sample 1 would typically be found at site:
    a. A
    b. B
    c. C

23. Sample 2 would typically be found at site:
    a. A
    b. B
    c. C

24. Sample 3 would typically be found at site:
    a. A
    b. B
    c. C



A 3D model of a landscape, its bedrock, and its groundwater.

25. This region is characterized by ___ topography.
    a. Karst

26. Does the low water table seen here promote or hinder the formation of this topography?
    a. Promote (allows water to drain and dissolve bedrock)

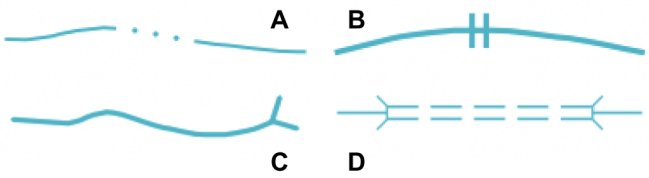
27. Select all of the following minerals that could be dominant in the bedrock here: (2)
    a. Quartz
    b. Dolomite
    c. Feldspar
    d. Halite (rock salt)
    e. Gypsum
    f. Calcite

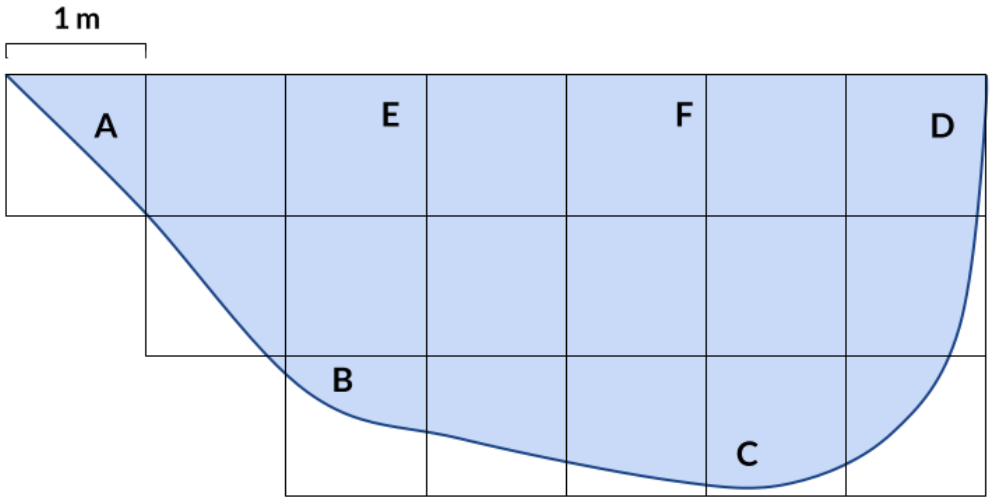
28. Why is pollution typically rare in groundwater? (2)
    a. Sediment acts as a filter to trap and remove particles like plastics and other pollutants. Water can move through small pores, but suspended particles typically can't.

29. How does this topography allow pollution to reach groundwater? (2)
    a. Karst topography allows better drainage, where water will go from the surface to the water table relatively easily through dissolved channels.

30. What type of stream is the one labeled A?
    a. Disappearing stream, losing stream, or influent stream

31. Which of the symbols in the following image would best represent feature A on a map?
    a. A
    b. B
    c. C
    d. D





A cross-section of a typical river.

32. This image is most likely depicting which of the following types of rivers?
    a. Straight
    b. Meandering
    c. Braided
    d. Anastomosing
    e. None of the above

33. At which point is river water likely moving the fastest? (2)
    a. A
    b. B
    c. C
    d. D
    e. E
    f. F

34. At which point(s) could saltation occur? Select all that apply.
    a. A
    b. B
    c. C
    d. D
    e. E
    f. F

35. At which point(s) would erosion likely be occurring? Select all that apply,
    a. A

b. B
c. C
d. D
e. E
f. F

36. At which point(s) would deposition likely be occurring? Select all that apply.
    a. A
    b. B
    c. C
    d. D
    e. E
    f. F

37. If the discharge of the river is measured to be 60 m^3/s, estimate the average velocity of the river in m/s. (2)
    a. 4 m/s (accept 3.5-4.5)

38. This river drains directly into a lake with a volume of 9 cubic kilometers. Using the discharge value from the previous question, estimate the residence time of the water in this lake. (One year is approximately 30 million seconds.) (2)
    a. 5 years (actual value is 4.76, accept anywhere from 4.5-5)

39. Why might the actual residence time be less than this value?
    a. Different water sources, e.g. rainfall, snow melt, another river, etc.

Section 2: Case Studies

40. The Aral Sea has rivers flowing into it, but no rivers flowing out. This makes it a(n) ___ lake.
    a. Endorheic

41. In the 1960s, the Soviet Union instituted a plan to use the Aral Sea to grow cash crops like cotton. However, this would have required huge amounts of fertilizer. If this fertilizer washed into the lake, it likely would have become a(n) ___ lake.
    a. Eutrophic

42. Why would pollution have been especially bad in a lake like the Aral Sea? (2)
    a. Because the lake has no natural outflows, pollution can't be removed - it simply stays there until it is absorbed or reacts to form something else.

43. To accomplish their goal, the Soviet Union diverted the Syr Darya and Amu Darya rivers for irrigation, removing the two main sources of the Aral Sea's water. Select all of the following events that occurred after this act: (2)
    a. The lake's volume decreased
    b. The lake's salinity decreased
    c. Evaporation increased
    d. Fish died rapidly

44. Which of the following changes would have primarily occurred to the region's climate?
    a. Cooling due to an increase in albedo
    b. Heating due to the release of water vapor, a greenhouse gas
    c. Smaller temperature range due to the moderating effect of airborne dust
    d. Larger temperature range due to the lack of the lake's moderating effect

45. In the 1980s and 1990s, Dike Kokaral was built out of local sand to contain the waters of the North Aral Sea, into which the Syr Darya was still flowing. However, this dam eventually failed twice for which of the following reasons?
    a. Fish and other animals burrowed into and destabilized the dam
    b. Sediment from the river built up, destabilizing the dam's foundations
    c. As water levels increased, water pressure increased until the dam broke
    d. Cooling increased the rate of ice wedging, breaking apart the dam

The Salton Sea has undergone a similar process as the Aral Sea; however, instead of being drained by human activity, it was created by human activity.

46. Several hundred years ago, the Colorado River flowed into the basin, creating a Salton Sea with a consistent water level that had a balance between what two types of water movement?

a. Runoff and evaporation (0.5 for each)

47. In 1580, the Colorado River changed to flow outside the valley, drying up the lake and creating what feature?
    a. Alluvial fan
    b. Desert pavement
    c. Bajada
    d. Playa

48. Which of the following types of sediment would have been most common in the valley after the lake dried up?
    a. Biogenous
    b. Terrigenous
    c. Evaporitic
    d. Anthropogenic

49. In 1905, the current Salton Sea was formed when the Colorado River was accidentally diverted into the valley via irrigation canals. This created a massive waterfall on dry desert sand and loose rock. Which of the following likely happened next?
    a. The waterfall disappeared rapidly, instead creating a smooth but steep gradient
    b. The waterfall eroded itself and rapidly receded upstream
    c. Falling water dug into the ground below it, allowing the sea to expand
    d. Sediment deposited at the top of the waterfall and blocked the river

50. While it was originally very successful, the sea began suffering from chronic algae blooms that caused massive die-offs. Why did this happen? (2)
    a. Irrigation indicates that there was significant farming in the region, which would have released fertilizer in runoff (1), giving algae the perfect nutrients to grow and take over the lake via eutrophication (1).

51. The current volume of the lake is 7.4 km^3 (much lower than when it began). The current salinity of the lake is roughly 44 kg/m^3, and the lake is estimated to deposit 4 million tons (4.06 billion kg) of salt each year. If evaporation stays constant, approximately when will the lake dry up completely? (2)
    a. 2102 (accept anywhere from 2090-2110)

Lake Vostok is a subglacial lake in Antarctica that is located beneath four kilometers of ice.

52. The lake's average temperature is approximately -3°C, well below the normal freezing point of water. Why is this water still liquid?
    a. The pressure of the ice above it lowers its freezing point (1), allowing it to remain liquid.

53. The ice covering the southern part of the lake is slightly thinner than the ice to the north. Which of the following best describes its water level?
    a. Even throughout the lake
    b. Higher at the northern end
    c. Higher at the southern end

54. Lake Vostok's salinity is highly debated, but suppose it has a significant amount of salt and other dissolved particles. If meltwater were to form at the northern end of the lake, what would it do? (2)
    a. Move south, because it is less dense and is forced upwards
    b. Move south, because it is denser and is forced downwards
    c. Stay in place, because it is less dense and is forced upwards
    d. Stay in place, because it is denser and is forced downwards

55. If Lake Vostok is saline, it is likely stratified year-round, since the ice sheet above it blocks any external forces. Which of the following best describes the lake?
    a. Meromictic
    b. Monomictic
    c. Polymictic
    d. Amictic

The Three Gorges Dam on the Yangtze River is the largest power station in the world, producing roughly 100 TWh of power annually. However, this dam has caused significant environmental impacts.

56. The purpose of the dam was to control damaging flooding of the river downstream by creating a reservoir behind it to collect and control water flow. Select all of the following that would occur **upstream** of the dam as a result of this:
    a. Flooding and rising water levels
    b. Landslides
    c. Lack of nutrients in water
    d. Reversal of river flow

57. Sediment deposition has been such a major problem for the dam that new gates had to be constructed to let sediment out. Why was sediment being deposited behind the dam instead of simply being carried through it? (2)
    a. The reservoir acts as a stagnant body of water, slowing down incoming rivers (1) and decreasing sediment capacity. As the water sits there waiting to be let out, sediment sinks to the bottom and collects there (1).

58. Forest cover has dropped to less than 10% in the Three Gorges region, compared to 20% in the 1950s. How could this worsen erosion and landslides? (2)

a. Tree roots (1) and other plants stabilize land, preventing erosion (1). Without them, the land is much more prone to erosion and mass wasting.

Herschel Island in Canada is facing significant impacts of climate change.

59. Would the melting of sea ice around the island increase or decrease coastal erosion? Why? (2)
    a. Increase (1) because it allows more active convection and surface currents in the ocean (1).

60. Select all of the following forms of mass wasting that could become more common as permafrost melts:
    a. Avalanches
    b. Rockfalls
    c. Slumps
    d. Solifluction

61. One common feature on Herschel Island is the presence of trees leaning in random directions due to the melting of permafrost. This feature is called a ___ forest.
    a. Drunken

62. In the formation of the previous feature, would ground level typically rise or sink? Why? (2)
    a. Sink (1), because ice is less dense than water and takes up more space (1).

63. Permafrost typically contains greenhouse gases like methane, which would create a ___ feedback loop that would ___ global warming.
    a. Positive; accelerate
    b. Positive; decelerate
    c. Negative; accelerate
    d. Negative; decelerate

In 2013, a flash flood in the Big Thompson River in Colorado devastated local communities. This was the river's largest flood since 1976, when an even larger flood killed 144 people.

64. The 2013 flood was the second largest flood in the last 157 years, when recording first began. What is the recurrence interval of this flood?
    a. 79 years

65. Severe flooding is typically associated with low ___, since if water can't enter the soil, it remains on the surface.
    a. Infiltration

66. Identify all of the following factors of the Big Thompson Canyon that would have exacerbated the severity of flooding: (2)
    - **a. Sparse vegetation**
    - **b. Silty soil**
    - c. Dry soil
    - **d. Steep gradient**

67. In their rebuilding efforts, crews opted to restore the river's natural path and floodplain instead of building levees to contain the river. If their goal is to prevent the severe, damaging flooding it has experienced, is this a good idea? (2)
    - a. No, it allows any flood to flow outward and destroy buildings
    - b. No, it allows more rainfall to reach the stream, making its floods worse
    - **c. Yes, it allows the stream to widen and contain the extra water**
    - d. Yes, the floodplain will absorb water before it can destroy buildings

68. An engineer wants to build a barrier near the river, but a geologist explains that if they simply let the river flood, it will do their work for them. What feature is the geologist referring to?
    - a. Natural levee

Crater Lake is a large lake situated within the caldera of Mount Mazama, a crater that formed when the mountain collapsed 7,700 years ago. Because of its high elevation, it has no natural tributaries.

69. Without human intervention, which of the following would best describe the ecology of the lake?
    - a. No wildlife
    - b. Some fish, but very few plants and algae
    - **c. Some underwater plants, but no fish or algae**
    - d. Abundant fish, plants, and algae

70. Crater Lake is known to have some of the clearest waters in the world. Select all of the following factors that contribute to this: (2)
    - **a. Little sediment due to the lack of tributaries**
    - b. Sterilization from volcanic heat
    - **c. Lack of nutrients for algae to form**
    - d. Strong year-round stratification

71. Which of the following best describes the balance of evaporation and precipitation in the lake? (2)
    - a. Evaporation is greater than precipitation
    - b. Precipitation is greater than evaporation
    - **c. Precipitation equals evaporation**
    - d. Impossible to tell with the given information

72. If the lake was smaller and there was more space on the inside of the crater, which of the following would best describe the drainage pattern of the mountain?
    a. Radial within the crater and dendritic outside it
    b. Dendritic within the crater and radial outside it
    c. Radial within the crater and centripetal outside it
    d. Centripetal within the crater and radial outside it

Month	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	Year
Record high °F (°C)	64 (18)	66 (19)	67 (19)	71 (22)	80 (27)	95 (35)	100 (38)	94 (34)	93 (34)	81 (27)	71 (22)	64 (18)	100 (38)
Average high °F (°C)	33.8 (1.0)	35.2 (1.8)	37.3 (2.9)	42.8 (6.0)	50.5 (10.3)	58.6 (14.8)	69.6 (20.9)	69.6 (20.9)	63.0 (17.2)	52.0 (11.1)	40.1 (4.5)	34.6 (1.4)	48.9 (9.4)
Daily mean °F (°C)	25.8 (−3.4)	26.8 (−2.9)	28.3 (−2.1)	32.6 (0.3)	39.4 (4.1)	46.3 (7.9)	55.2 (12.9)	55.1 (12.8)	49.7 (9.8)	41.2 (5.1)	31.8 (−0.1)	26.8 (−2.9)	38.3 (3.5)
Average low °F (°C)	17.7 (−7.9)	18.3 (−7.6)	19.2 (−7.1)	22.3 (−5.4)	28.2 (−2.1)	33.9 (1.1)	40.7 (4.8)	40.6 (4.8)	36.3 (2.4)	30.3 (−0.9)	23.4 (−4.8)	19.0 (−7.2)	27.5 (−2.5)
Record low °F (°C)	−21 (−29)	−18 (−28)	−7 (−22)	−3 (−19)	5 (−15)	10 (−12)	18 (−8)	23 (−5)	16 (−9)	3 (−16)	−7 (−22)	−18 (−28)	−21 (−29)
Average precipitation inches (mm)	10.20 (259)	7.93 (201)	7.74 (197)	4.96 (126)	3.22 (82)	2.31 (59)	0.75 (19)	0.95 (24)	2.08 (53)	1.94 (49)	9.25 (235)	11.14 (283)	62.47 (1,587)
Average snowfall inches (cm)	96.2 (244)	82.0 (208)	82.2 (209)	45.3 (115)	19.7 (50)	3.9 (9.9)	0.2 (0.51)	0.1 (0.25)	2.8 (7.1)	21.4 (54)	60.8 (154)	90.4 (230)	505 (1,281.76)

73. Which of the following best describes the seasonal stratification of Crater Lake? (2)
    a. Well-mixed year-round
    b. Well-mixed in summer and stratified in winter
    c. Stratified in summer and well-mixed in winter
    d. Stratified year-round

74. Assuming evaporation stays constant year-round, in what month is the lake's water level usually the lowest?
    a. October

75. Assuming evaporation stays constant year-round, in what month is the lake's water level usually the highest?
    a. March or April (accept either)